

Assignment1:

Without TL;DR:

The screenshot shows a web browser with multiple tabs. The active tab is ChatGPT, displaying a conversation with the user's prompt "Explain me BRUNO Tool". The response explains that Bruno is an open-source API testing tool, lists its official website and documentation, and highlights its key features: Git-friendly, Offline-first, and Privacy-focused. It also mentions that Bruno stores data locally. A "Simple Explanation" section is partially visible at the bottom.

ChatGPT

Explain me BRUNO Tool

Bruno is an **open-source API testing tool** used by developers to test and work with APIs (REST, GraphQL, gRPC, SOAP, WebSockets, etc.). It is often considered an alternative to tools like Postman and Insomnia.

Official website: [Bruno](#)
Documentation: [Bruno Docs](#)

What makes Bruno special?

The main idea behind Bruno is:

- **Git-friendly**
- **Offline-first**
- **Privacy-focused**

Unlike many API tools that save collections to the cloud, Bruno stores everything as local files on your computer. 🍌 Bruno Docs +1

Simple Explanation

Imagine you are building a backend API like:

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This screenshot shows the continuation of the ChatGPT response. It provides a sample HTTP request in a code block and lists the tasks Bruno helps with: sending requests, checking responses, testing authentication, and debugging APIs. It then introduces the "Core Features of Bruno" section, starting with "1. API Testing".

Simple Explanation

Imagine you are building a backend API like:

```
http
GET /users
POST /login
PUT /profile
```

You need a tool to:

- send requests
- check responses
- test authentication
- debug APIs

Bruno helps you do this easily.

Core Features of Bruno

1. API Testing

You can send:

- **GFT requests**

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PUT request

- PUT requests
- DELETE requests

Example:

```
http
GET https://api.example.com/users
```

Bruno shows:

- response body
- status code
- headers
- timing

2. Supports Multiple API Types

Bruno supports:

- REST APIs
- GraphQL
- SOAP
- gRPC

+

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3. Collections

You can group requests into collections.

Example:

```
User APIs
├── Login
├── Register
├── Get Profile
└── Update Profile
```

4. Git Integration (Very Important)

This is Bruno's biggest selling point.

Bruno stores collections as plain text `.bru` files.

Example:

+

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login.bru
users.bru
profile.bru

Because they are plain files:

- you can commit them to Git
- track changes
- review them in pull requests
- collaborate with teams easily

This is why developers love Bruno. [toolstac.com](#) +1

Why Developers Prefer Bruno Over Postman

Feature	Bruno	Postman
Open Source	✓	✗
Offline First	✓	Partial
Git Friendly	✓ Native	Limited
Cloud Required	✗	Usually Yes

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Login Required

Often Yes

toolstac.com +1

Example Workflow

Suppose you built a login API.

Step 1 — Create Request

```
http
POST /login
```

Step 2 — Add JSON Body

```
JSON
{
  "email": "test@gmail.com",
  "password": "123456"
}
```

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JSON

Step 3 — Send Request

Bruno returns:

```
JSON
{
  "token": "abc123"
}
```

Step 4 — Use Token in Other APIs

You can store variables like:

```
{{token}}
```

and reuse them.

Bruno File Example (.bru)

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Bruno File Example (.bru)

A Bruno request is saved like this:

```
meta {
  name: Get Users
  type: http
  seq: 1
}

get {
  url: https://api.example.com/users
}
```

Human-readable and easy for Git tracking.

Scripting & Automation

Bruno also supports:

- pre-request scripts
- test scripts
- environment variables

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Bruno CLI

Bruno provides a CLI tool for running collections from terminal.

Example:

```
Bash
bru run collection-name
```

Useful in:

- Jenkins
- GitHub Actions
- CI/CD pipelines

Advantages

- Fast
- Lightweight
- Open source
- Works offline

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Limitations

Compared to Postman:

- smaller ecosystem
- fewer enterprise collaboration features
- newer community
- some advanced workflows are still evolving

But it is improving rapidly. +1

Who Should Use Bruno?

Bruno is excellent for:

- Backend developers
- Full-stack developers
- DevOps engineers
- QA testers
- Teams using Git heavily

Especially good if you:

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Bruno Tool Overview

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Installation

You can download Bruno from:

[Download Bruno](#)

Available for:

- Windows
- macOS
- Linux

In One Line

Bruno is a modern, open-source, Git-friendly API testing tool designed as a lightweight and privacy-focused alternative to Postman.

Citations

Bruno Docs

What is Bruno? - Bruno Docs

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TL;DR Explain me BRUNO Tool

Bruno is an **open-source API testing and development tool** — basically a modern alternative to tools like Postman or Insomnia.

Official site: [Bruno](#)

In simple terms

Developers use Bruno to:

- Send API requests (GET, POST, PUT, etc.)
- Test backend APIs
- Debug authentication/tokens
- Save API collections
- Automate API testing

Example:

If your backend exposes:

↗ http

GET /users

POST /login

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```
http
POST /login
```

You can use Bruno to test these endpoints without writing frontend code.

Why Bruno became popular

1. Git-friendly (its biggest feature)

Unlike Postman, Bruno stores collections as **plain text files** directly on your computer.

That means:

- You can commit API collections into Git
- Teams can review API changes like code
- No weird cloud sync conflicts

Example:

```
Bash
project/
├── backend/
└── frontend/
    ├── api-tests/
    └── login.bru
```

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```
Bash
project/
├── backend/
└── frontend/
    ├── api-tests/
    └── login.bru
```

This makes collaboration much cleaner. 🧑 Bruno +1

2. Offline-first

Bruno works locally.

No mandatory:

- login
- account
- cloud sync

Your API secrets and tokens stay on your machine. 🧑 Bruno +1

3. Test-first

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3. Lightweight

Compared to Postman:

- faster startup
- less RAM usage
- simpler UI
- less "enterprise bloat"

What a Bruno request looks like

Bruno stores requests in `.bru` files.

Example:

```
bru
meta {
  name: Get Users
  type: http
}

get {
  url: https://api.example.com/users
}
```

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This is human-readable and Git-diff friendly.

Main Features

API Types Supported

Bruno supports:

- REST APIs
- GraphQL
- gRPC
- WebSockets (newer versions)

Tool Question +1

Environment Variables

You can store:

- API keys
- JWT tokens
- base URLs

Example:

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- API keys
- JWT tokens
- base URLs

Example:

```
env
BASE_URL=https://dev-api.com
TOKEN=abc123
```

Scripting & Testing

You can write:

- pre-request scripts
- post-response tests

Example:

```
JavaScript
test("status should be 200", function() {
  expect(res.status).toEqual(200);
});
```

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CLI Support

Bruno also has a CLI tool for:

- CI/CD pipelines
- automated API testing
- GitHub Actions

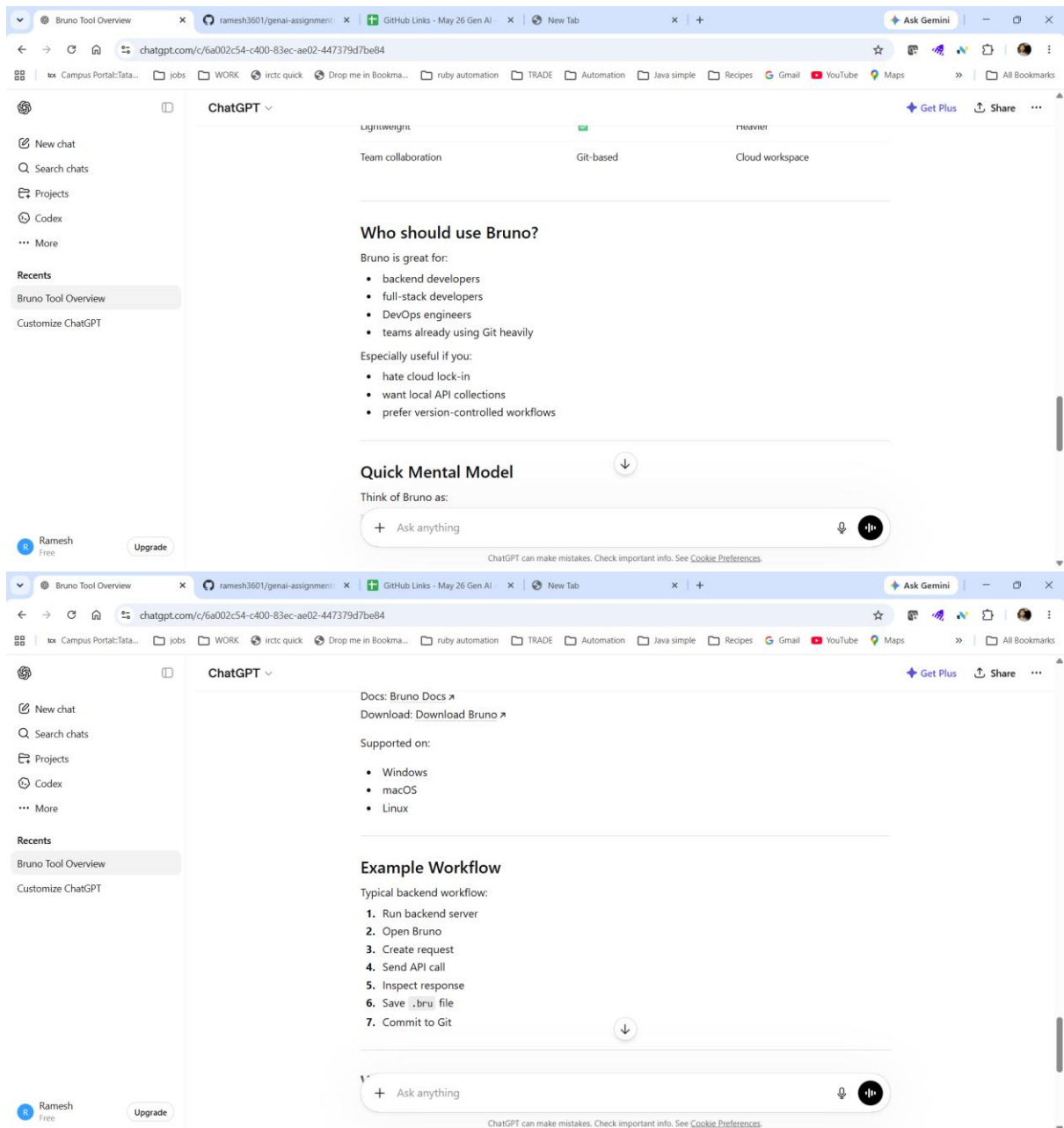
Bruno Docs

Bruno vs Postman

Feature	Bruno	Postman
Open-source	✓	Partial
Offline-first	✓	✗
Git-native	✓	Limited
Cloud account required	✗	Usually yes

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ELI 15 explain me Bruno tool

Imagine you built a game backend.

Your app has APIs like:

http

POST /login

GET /profile

POST /buy-item

Now you need a way to:

- send requests to those APIs
- see responses
- debug errors
- test tokens/auth

That's what Bruno does. Bruno +1

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ELI15 Version

Think of Bruno as:

A "remote control" for talking to servers.

Instead of building a full frontend app just to test your backend, you directly send requests using Bruno.

Real-life analogy

Suppose you ordered food using Swiggy.

Your app internally sends requests like:

JSON

POST /place-order

{

"item": "Pizza"

}

Normally:

- User clicks button

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- User clicks button
- Frontend sends request
- Backend responds

With Bruno:

- YOU manually send the request
- YOU inspect the backend response

So Bruno helps developers test APIs faster.

Why developers like Bruno

Older tools like Postman became:

- cloud-heavy
- account-heavy
- slower
- more enterprise-focused

Bruno said:

"What if API testing was simple, local, and Git-frie...?"

That became its main selling point. Bruno +1

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The BIG idea of Bruno

Everything is just files

Bruno stores API requests as plain files on your computer.

Example:

```
Bash
api-tests/
├── login.bru
├── users.bru
└── payments.bru
```

This is HUGE because:

- Git can track changes
- teammates can collaborate easily
- requests become version-controlled

That's Bruno's "Git-native" idea. Bruno +1

What is .bru ?

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Bruno invented a simple text format.

Example:

```
bru
meta {
  name: Login
}

post {
  url: https://api.app.com/login
}
```

This file literally describes the API request.

Human-readable.

Easy to share.

Easy to Git commit.

How developers use it

Typical workflow:

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How developers use it

Typical workflow:

Step 1

Backend engineer creates API:

```
http
POST /login
```

Step 2

Open Bruno

Step 3

Send request:

```
JSON
{
```

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```
JSON
{
  "email": "john@test.com",
  "password": "123"
}
```

Step 4

Backend responds:

```
JSON
{
  "token": "abc123"
}
```

Step 5

Developer checks:

- Did it work?

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- Correct status code?
- Any bugs?
- Wrong JSON?
- Auth issue?

Why Git integration matters

This is the coolest part.

With Postman:

- collections usually live in cloud workspaces

With Bruno:

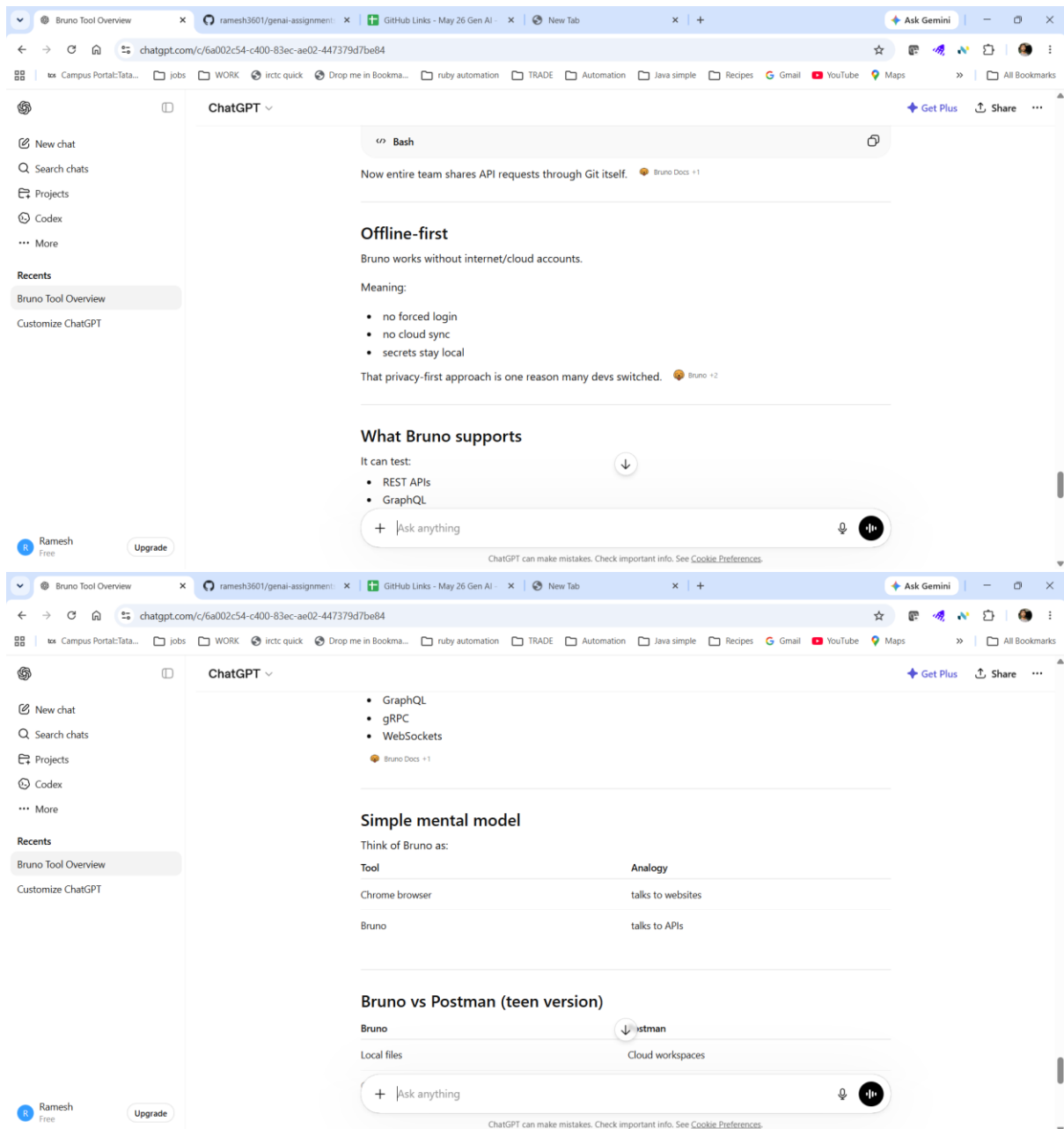
- collections live inside your project repo

Example:

```
Bash
my-project/
├── backend/
├── frontend/
└── bruno/
```

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Assignment 2:

2.a Automation framework architecture:

Context:

Experienced Functional tester but new to API Testing

Instruction:

Design a API Framework architecture using Bruno Tool and validate the request and response given below.

[MANDATORY] Generate only one testdata set and apply in the below created framework.

[MANDATORY] Application id field value should be AlphaNumeric with 9 digit (Example: AO124567).

[MANDATORY] Staging identifier field value should be Number only with 10 digit(Example:0001234567).

[MANDATORY] Trigger a GET request send with application id

[STRICT] The above GET request should return all the values (Title,Firstname,Middlename,Lastname,DOB,Email address,CurrentAddress, Postalcode, Nationality,Monhtly Income, Monthly spending Amount,No of Dependents,Staging identifier,Application id).

[STRICT] GET request should have Staging identifier and it should unique.

[MANDATORY] Save the above response json of the GET request.

[STRICT] Staging table named as Customer details should have Title, Firstname, Middlename, Lastname, DOB, Gender, Email address, Postalcode, Nationality,Staging identifier.

[STRICT] Staging identifier value of GET response and POST request should be same.

[MANDATORY] Trigger a POST request by passing above generated Staging identifier value.

[STRICT] GET request and POST request should be success.

[MANDATORY] the POST request should return staging table Customer details values in json format as a response.

[MANDATORY] Save the json response of the POST request.

[STRICT] Validate the saved json response of POST request against saved json response of GET request for the fields -Title,Firstname,Middlename,Lastname, DOB,Email address,Postalcode,Nationality.

[STRICT] While assertion, Middlename should mismatch.

[STRICT] All field values -Firstname, LastName, Middlename, Emailid,Psotalcode should contain Indian names.

[MANDATORY] Ask me 7 multiple choice questions one by one.

[STRICT] Once after answering one question goto next one by one.

[CRITICAL] Generate API FRAMEWORK using BRUNO tool after answering all questions with above GET request,POST request, GET Response,POST response.

API Framework Design

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Bruno API Framework Architecture

Based on your selections:

- Authentication → Bearer Token
- Environment → Dev + QA
- Validation → Status + Schema + Field Validation
- Execution → Collection Runner
- Reports → HTML + JSON
- Assertions → Chai Assertions
- Framework Structure → Request-wise folders

Recommended Bruno Framework Structure

```
Bruno_API_Framework/  
├── environments/  
│   ├── dev.bru  
│   └── qa.bru  
├── collections/  
│   └── GET_Application_Details/  
│       └── get_application_details.bru  
└──
```

Ask anything

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API Framework Design

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Recommended Bruno Framework Structure

```
Bruno_API_Framework/  
├── environments/  
│   ├── dev.bru  
│   └── qa.bru  
├── collections/  
│   ├── GET_Application_Details/  
│   │   ├── get_application_details.bru  
│   │   ├── get_response.json  
│   │   └── tests.js  
│   ├── POST_Customer_Details/  
│   │   ├── post_customer_details.bru  
│   │   ├── post_response.json  
│   │   └── tests.js  
│   └── schemas/  
│       ├── get_response_schema.json  
│       └── post_response_schema.json  
├── testdata/  
│   └── customer_testdata.json  
├── reports/  
│   ├── html/  
│   └── json/  
└── README.md
```

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Single Mandatory Test Data Set

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JSON

{
 "applicationId": "A01245678",
 "stagingIdentifier": "0001234567",
 "title": "Mr",
 "firstName": "Arjun",
 "middleName": "Kumar",
 "lastName": "Raman",
 "dob": "1995-08-15",
 "gender": "Male",
 "emailAddress": "arjun.raman@testmail.in",
 "currentAddress": "12 Gandhi Street Chennai",
 "postalCode": "600001",
 "nationality": "Indian",
 "monthlyIncome": 85000,
 "monthlySpendAmount": 25000,
 "noOfDependents": 2
}

Mandatory Validation Rules Covered

Application ID Validation

JavaScript

const appIdRegex = /^[A-Za-z0-9]{9}\$/;

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Mandatory Validation Rules Covered

Application ID Validation

JavaScript

```
const appIdRegex = /^[A-Za-z0-9]{9}$/;
pm.expect(applicationId).to.match(appIdRegex);
```

Valid Example:

A01245678

Staging Identifier Validation

JavaScript

```
const stagingRegex = /^[0-9]{10}$/;
pm.expect(stagingIdentifier).to.match(stagingRegex);
```

Valid Example:

0001234567

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GET Request Design

Endpoint

http

```
GET /applications/A01245678
```

Headers

http

```
Authorization: Bearer {{token}}
Content-Type: application/json
```

Expected GET Response

JSON

```
{
  "title": "He",
  "firstName": "Arjun",
  "middleName": "Kumar",
  "lastName": "Raman",
  "dob": "1995-08-15",
  "emailAddress": "arjun.raman@testmail.in",
  "currentAddress": "12 Gandhi Street Chennai",
  "postalCode": "600001",
  "nationality": "Indian",
```

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Expected GET Response

```
JSON
{
  "title": "Mr",
  "firstName": "Arjun",
  "middleName": "Kumar",
  "lastName": "Raman",
  "dob": "1995-08-15",
  "emailAddress": "arjun.raman@testmail.in",
  "currentAddress": "12 Gandhi Street Chennai",
  "postalCode": "600001",
  "nationality": "Indian",
  "monthlyIncome": 85000,
  "monthlySpendAmount": 25000,
  "numberOfDependents": 2,
  "stagingIdentifier": "0001234567",
  "applicationId": "A01345678"
}
```

GET Request Assertions

```
JavaScript
const chai = require('chai');
const expect = chai.expect;
```

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GET Request Assertions

```
JavaScript
const chai = require('chai');
const expect = chai.expect;

const response = res.getBody();

expect(res.getStatus()).to.equal(200);

expect(response.applicationId).to.match(/^[A-Za-z0-9]{9}$/);
expect(response.stagingIdentifier).to.match(/^[0-9]{10}$/);
expect(response.firstName).to.equal("Arjun");
expect(response.lastName).to.equal("Raman");
expect(response.middleName).to.equal("Kumar");
expect(response.emailAddress).to.contain(".in");

bru.setVar("savedStagingIdentifier", response.stagingIdentifier);
bru.setVar("savedGetResponse", JSON.stringify(response));
```

Save GET Response JSON

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Save GET Response JSON

File:

collections/GET_Application_Details/get_response.json

Saved Content:

JSON

```
{
  "title": "Mr.",
  "firstName": "Arjun",
  "middleName": "Kumar",
  "lastName": "Raman",
  "dob": "1995-08-15",
  "emailAddress": "arjun.ram@testmail.in",
  "currentAddress": "12 Gandhi Street Chennai",
  "postalCode": "600001",
  "nationality": "Indian",
  "monthlyIncome": 85000,
  "monthlySpendingAmount": 25000,
  "noOfDependents": 2,
  "stagingIdentifier": "0001234567",
  "applicationId": "A01245678"
}
```

Staging Table — Customer Details

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Staging Table — Customer Details

Field

Title

Firstname

Middlename

Lastname

DOB

Gender

Email Address

Postalcode

Nationality

Staging Identifier

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POST Request Design

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Endpoint

http

POST /customer/details

Headers

http

Authorization: Bearer {{token}}

Content-Type: application/json

POST Request Body

JSON

{

"stagingIdentifier": "0001234567"

}

Expected POST Response

JSON

{

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Expected POST Response

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JSON

{

"title": "Mr",

"firstName": "Arjun",

"middleName": "Kumar",

"lastName": "Raman",

"dob": "1995-08-15",

"gender": "Male",

"emailAddress": "arjun.raman@testmail.in",

"postalCode": "600001",

"nationality": "Indian",

"stagingIdentifier": "0001234567"

}

Important Mandatory Negative Validation

Middlename Should Mismatch

GET Response:

JSON

"middleName": "Kumar"

POST Response:

JSON

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POST Request Assertions

```
JavaScript
const chai = require('chai');
const expect = chai.expect;

const postResponse = res.getBody();

expect(res.getStatus()).to.equal(200);
expect(postResponse.stagingIdentifier
.to.equal(bvu.getVar("savedStagingIdentifier"));

const getResponse =
JSON.parse(bvu.getVar("savedGetResponse"));

expect(postResponse.title)
.to.equal(getResponse.title);

expect(postResponse.firstName)
.to.equal(getResponse.firstName);

expect(postResponse.lastName)
.to.equal(getResponse.lastName);

expect(postResponse.dob)
.to.equal(getResponse.dob);

expect(postResponse.emailAddress)
.to.equal(getResponse.emailAddress);

expect(postResponse.postalCode)
.to.equal(getResponse.postalCode);
```

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```
JavaScript
expect(postResponse.middleName)
.to.not.equal(getResponse.middleName);
```

Save POST Response JSON

File:

collections/POST_Customer_Details/post_response.json

Saved Content:

```
JSON
{
  "title": "He",
  "firstName": "Arjun",
  "middleName": "Kumar",
  "lastName": "Raman",
  "dob": "1995-08-15",
  "gender": "Male",
  "emailAddress": "arjun.raman@testmail.in",
  "postalCode": "600001",
  "nationality": "Indian",
  "stagingIdentifier": "0001234567"
}
```

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Schema Validation Example

GET Schema

```
JSON
{
  "type": "object",
  "required": [
    "title",
    "firstName",
    "middleName",
    "lastName",
    "dob",
    "emailAddress",
    "postalCode",
    "nationality",
    "stagingIdentifier",
    "applicationId"
  ]
}
```

Execution Flow

1. Load Environment
2. Execute GET Request using Application ID
3. Validate GET Response

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Execution Flow

1. Load Environment
2. Execute GET Request using Application ID
3. Validate GET Response
4. Save GET Response JSON
5. Extract Staging Identifier
6. Execute POST Request using same Staging Identifier
7. Validate POST Response
8. Save POST Response JSON
9. Compare GET vs POST Responses
10. Verify MiddleName mismatch
11. Generate HTML + JSON Reports

Bruno Environment Example

dev.bru

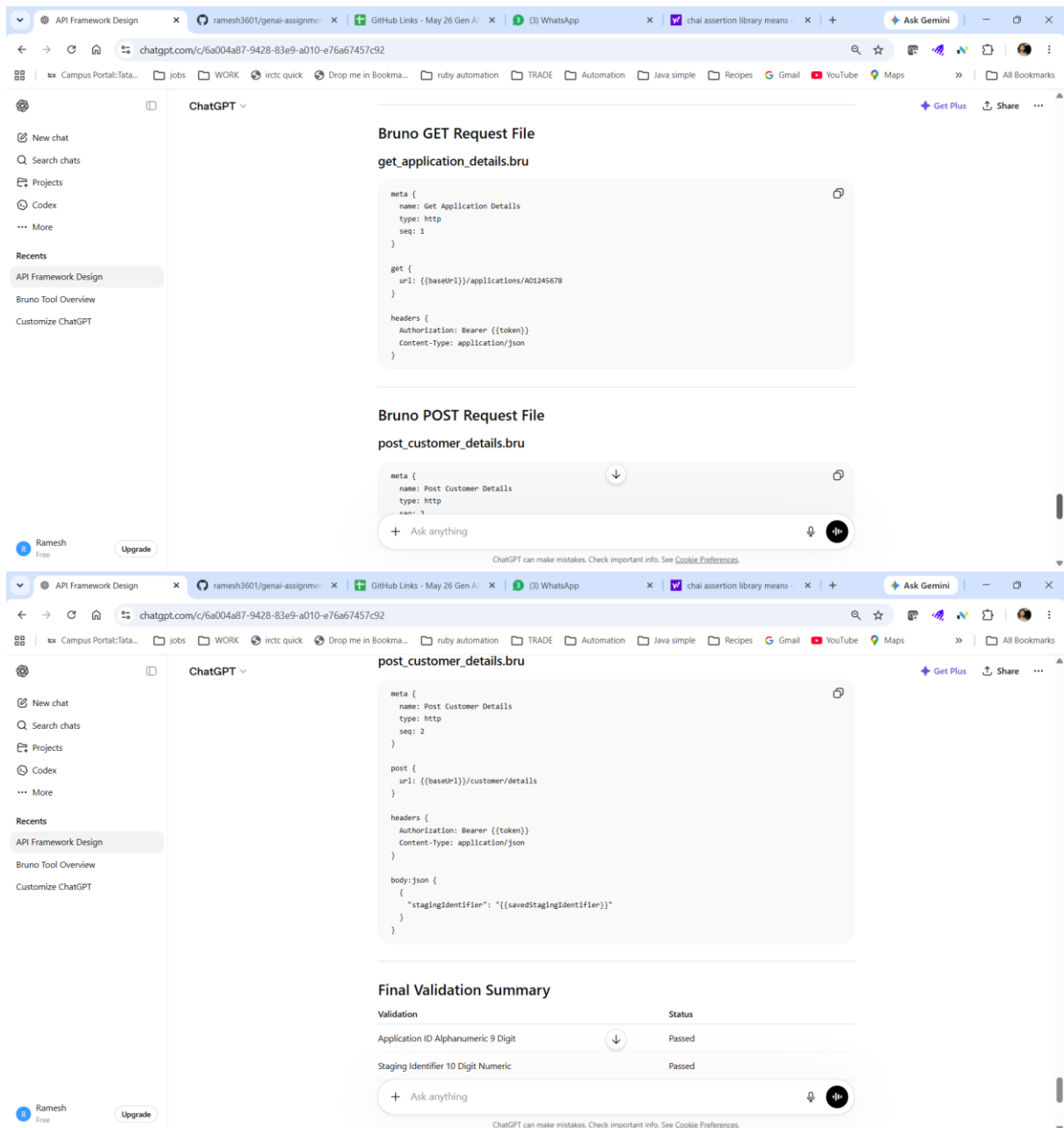
```
vars {
  baseUrl: https://dev-api.test.com
  token: dev_token_123
}
```

qa.bru

```
vars {
  baseUrl: https://qa-api.test.com
  token: qa_token_123
}
```

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2.B Creating Test Strategy document for API testing

Context: Experienced Functional tester, but new to API testing.

Instruction: Create a Test Strategy document for Automation API testing.

[MANDATORY] Ask me 10 multiple choice questions one by one.

[STRICT] Once after answering one question goto next one by one.

[CRITICAL] Generate Test Strategy document for API testing after answering all questions.

API Test Strategy Q&A

Context:
Experienced Functional tester, but new to API testing.

Instruction:
Create a Test Strategy document for Automation API testing.
[MANDATORY] Ask me 10 multiple choice questions one by one.
[STRICT] Once after answering one question goto next one by one.
[CRITICAL] Generate Test Strategy document for API testing after answering all questions.

Question 1 of 10
What is the **primary type of API** you want to automate testing for?

A. REST API
B. SOAP API
C. GraphQL API
D. Mixed APIs (REST + SOAP/GraphQL)

↓

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API Test Strategy Q&A

Question 2 of 10
Which **automation tool/framework** do you prefer for API testing?

A. Postman + Newman
B. REST Assured (Java)
C. Karate Framework
D. Playwright / Cypress API
E. Not decided yet

BRUNO TOOL

Question 3 of 10
What is the **main objective** of your API automation?

A. Functional validation only
B. Regression testing
C. Integration/end-to-end validation
D. Smoke/Sanity testing
E. All of the above

↓

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Question 4 of 10
Which environment(s) will the API automation run against?

A. Dev only
B. QA/Test only
C. Staging/UAT only
D. Multiple environments (Dev, QA, UAT, Prod-like)
E. Not finalized yet

B

Question 5 of 10
How will the APIs be authenticated?

A. No authentication
B. Basic Authentication
C. OAuth 2.0 / Bearer Token
D. API Key
E. Mixed authentication methods

C

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Question 6 of 10
What level of reporting do you expect from the automation framework?

A. Console logs only
B. HTML reports
C. Detailed reports with request/response logs
D. Dashboard integration (Jenkins, Azure DevOps, etc.)
E. All of the above

C

Question 7 of 10
How do you want test data to be managed?

A. Hardcoded test data
B. External JSON/XML files
C. Database-driven test data
D. Dynamic/generated test data
E. Combination of external files and dynamic data generation

E

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API Test Strategy Q&A

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Question 8 of 10

How should the API automation be integrated into execution pipelines?

A. Manual execution only

B. Scheduled execution

C. CI/CD integration (Jenkins/GitHub Actions/Azure DevOps)

D. Both manual and CI/CD execution

E. Not decided yet

D

Question 9 of 10

What type of validations are most important in your API tests?

A. Status code validation only

B. Response body/schema validation

C. Database validation

D. Performance/response time validation

E. Combination of all major validations

A,B,C

Ask anything

API Test Strategy Q&A

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Question 10 of 10

What level of framework maintainability do you expect?

A. Simple scripts for quick automation

B. Modular reusable framework

C. Enterprise-level scalable framework

D. Keyword/data-driven framework

E. Hybrid modular + data-driven framework

B

Test Strategy Document – API Automation Testing

1. Introduction

This document defines the Test Strategy for API Automation Testing of the application APIs using the Bruno API testing tool. The strategy focuses on validating integration workflows, ensuring API reliability, improving regression coverage, and supporting continuous testing activities.

The user is experienced in Functional Testing and is transitioning into API Automation Testing; therefore, this strategy emphasizes maintainable, modular, and scalable automation practices.

2. Objectives

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Test Strategy Document – API Automation Testing

1. Introduction

This document defines the Test Strategy for API Automation Testing of the application APIs using the Bruno API testing tool. The strategy focuses on validating integration workflows, ensuring API reliability, improving regression coverage, and supporting continuous testing activities.

The user is experienced in Functional Testing and is transitioning into API Automation Testing; therefore, this strategy emphasizes maintainable, modular, and scalable automation practices.

2. Objectives

The key objectives of API Automation Testing are:

- Validate API integration workflows
- Ensure APIs function correctly across QA environments
- Detect integration defects early
- Improve regression testing efficiency
- Validate request and response behaviors
- Verify API data integrity with database validation
- Provide faster feedback during releases

3. Scope of Testing

In Scope

- REST API validation
- Request and response verification

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- REST API validation
- Request and response verification
- Status code validation
- Response body validation
- Schema validation
- Database validation
- Authentication validation using OAuth 2.0 / Bearer Tokens
- Regression suite execution
- Integration workflow validation
- Error handling validation
- Environment-based execution (QA)

Out of Scope

- UI Testing
- Performance/Load Testing
- Security/Penetration Testing
- Production environment execution

4. Automation Tool & Technology Stack

Component	Selected Tool/Technology
API Automation Tool	Bruno
API Type	REST API
Authentication	OAuth 2.0 / Bearer Token
Environment	QA Environment
Regression	Parallel UTM: Request with Request Response Log

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API Test Strategy Q&A

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API typeREST API

AuthenticationOAuth 2.0 / Bearer Token

EnvironmentQA Environment

ReportingDetailed HTML Reports with Request/Response Logs

Execution ModeManual + CI/CD

Framework TypeModular Reusable Framework

5. Test Approach

5.1 Automation Approach

A modular reusable automation framework will be implemented to improve:

- Reusability
- Maintainability
- Scalability
- Easy onboarding for new APIs

The framework will support:

- Collection-based execution
- Environment configuration handling
- Reusable authentication setup
- Shared utility scripts
- Centralized test data management

5.2 API Validation Coverage

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5.2 API Validation Coverage

The automation suite will validate:

Functional Validation

- HTTP status codes
- Request payload validation
- Response payload validation
- Mandatory field validation
- Business rule validation

Schema Validation

- JSON schema structure
- Data type validation
- Required attribute validation

Database Validation

- Backend data persistence verification
- Data consistency checks
- API to DB reconciliation

Negative Testing

- Invalid requests
- Unauthorized access
- Missing mandatory fields
- Invalid tokens
- Boundary validations

6. Test Data Management

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6. Test Data Management

The framework will use a combination of:

- External JSON files
- Dynamic test data generation

Strategy

- Environment-specific test data separation
- Reusable payload templates
- Dynamic token generation
- Parameterized requests
- Secure handling of sensitive data

7. Environment Strategy

Target Environment

- QA/Test Environment

Environment Requirements

- Stable API endpoints
- Valid OAuth credentials
- Database access for validation
- Environment configuration files
- Mock/stub dependencies if required

8. Authentication Strategy

The APIs use OAuth 2.0 / Bearer Token authentication.

Ask anything

Ramesh

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8. Authentication Strategy

The APIs use OAuth 2.0 / Bearer Token authentication.

Automation Handling

- Dynamic token generation before execution
- Secure token storage
- Token refresh handling
- Reusable authentication module

9. Framework Design

Proposed Folder Structure

```
API-Automation/  
├── collections/  
├── environments/  
├── test-data/  
├── utilities/  
├── auth/  
├── reports/  
├── database/  
└── configs/
```

Framework Features

- Reusable request modules
- Environment switching
- Centralized configuration
- Logging support

Ask anything

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Logging support

Detailed execution reports

CI/CD compatibility

10. Reporting Strategy

Detailed reports will include:

- Test execution summary
- Passed/failed test cases
- Request payloads
- Response payloads
- Error messages
- Execution timestamps

Reporting Benefits

- Faster debugging
- Improved defect analysis
- Better traceability

11. CI/CD Strategy

The automation suite will support:

- Manual execution
- CI/CD pipeline execution

Pipeline Integration Possibilities

- Jenkins
- GitHub Actions

Ask anything

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Pipeline Integration Possibilities

- Jenkins
- GitHub Actions
- Azure DevOps

Execution Triggers

- Scheduled execution
- On-demand execution
- Build-based execution

12. Entry Criteria

API Automation Testing can start when:

- APIs are stable
- API documentation is available
- Test environment is accessible
- Authentication credentials are available
- Test data is prepared

13. Exit Criteria

Testing will be considered complete when:

- All critical APIs are automated
- Regression suite passes successfully
- High severity defects are closed
- Reports are generated successfully
- Stakeholder sign-off is completed

Ask anything

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13. Exit Criteria

Testing will be considered complete when:

- All critical APIs are automated
- Regression suite passes successfully
- High severity defects are closed
- Reports are generated successfully
- Stakeholder sign-off is completed

14. Risk Assessment

Risk	Mitigation
Unstable APIs	Coordinate with development team
Token expiration issues	Implement dynamic token handling
Environment downtime	Use retry mechanisms
Test data dependency	Use dynamic data generation
Frequent API changes	Modular framework design

15. Deliverables

The following deliverables will be produced:

- API Automation Framework
- Automated Test Scripts
- Test Data Files
- Execution Reports

Ask anything

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THE FOLLOWING DELIVERABLES WILL BE PRODUCED:

- API Automation Framework
- Automated Test Scripts
- Test Data Files
- Execution Reports
- Defect Reports
- CI/CD Integration Setup
- Test Strategy Document

16. Best Practices

- Use reusable request templates
- Keep test data externalized
- Avoid hardcoded values
- Implement proper assertions
- Maintain clear naming conventions
- Enable detailed logging
- Execute regression suites regularly
- Version control all automation assets

17. Conclusion

This API Automation Testing Strategy provides a structured and scalable approach for validating REST APIs using Bruno. The strategy focuses on integration validation, maintainability, reusable automation architecture, and detailed reporting to ensure reliable API quality assurance within the QA environment.

Ask anything